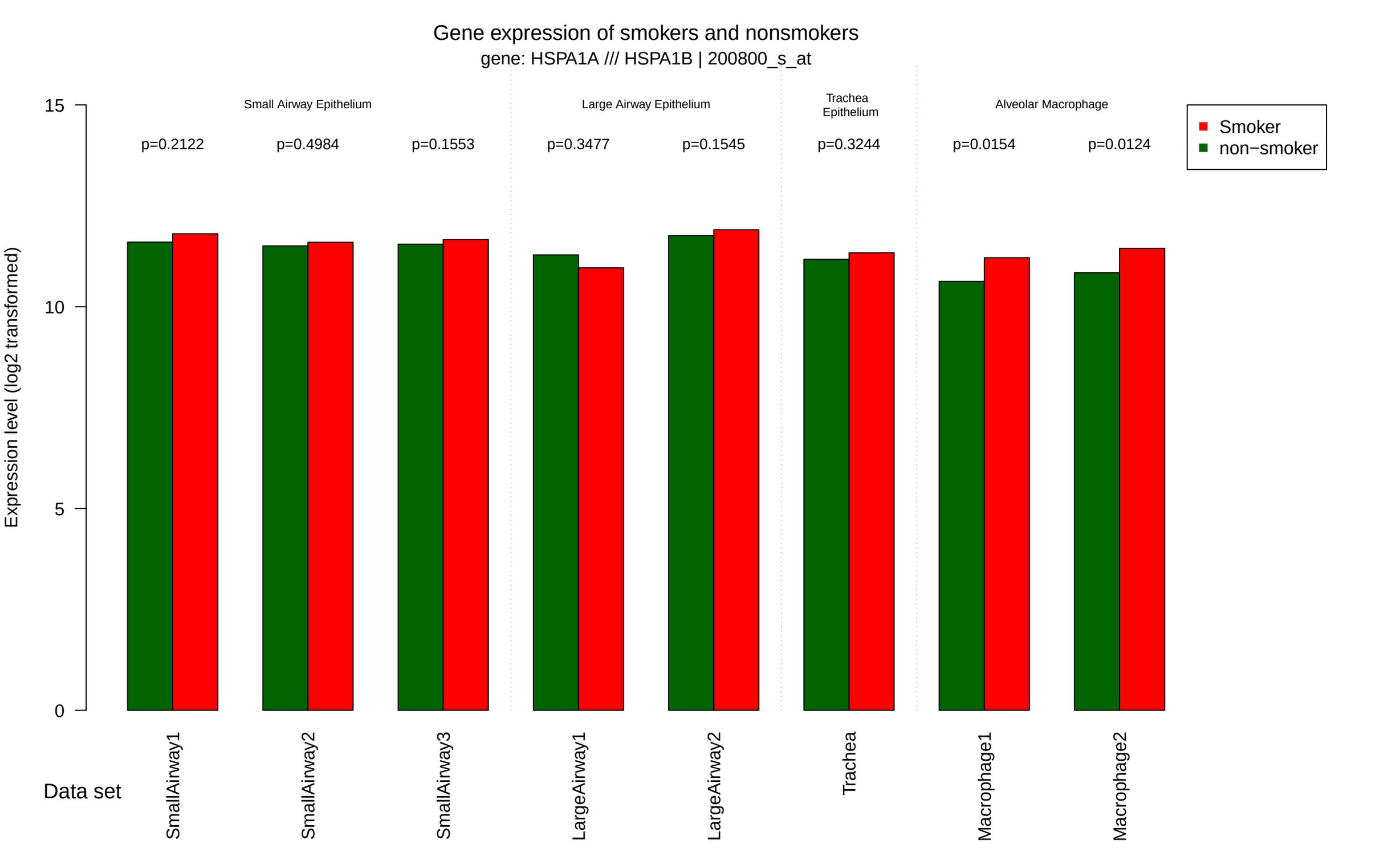
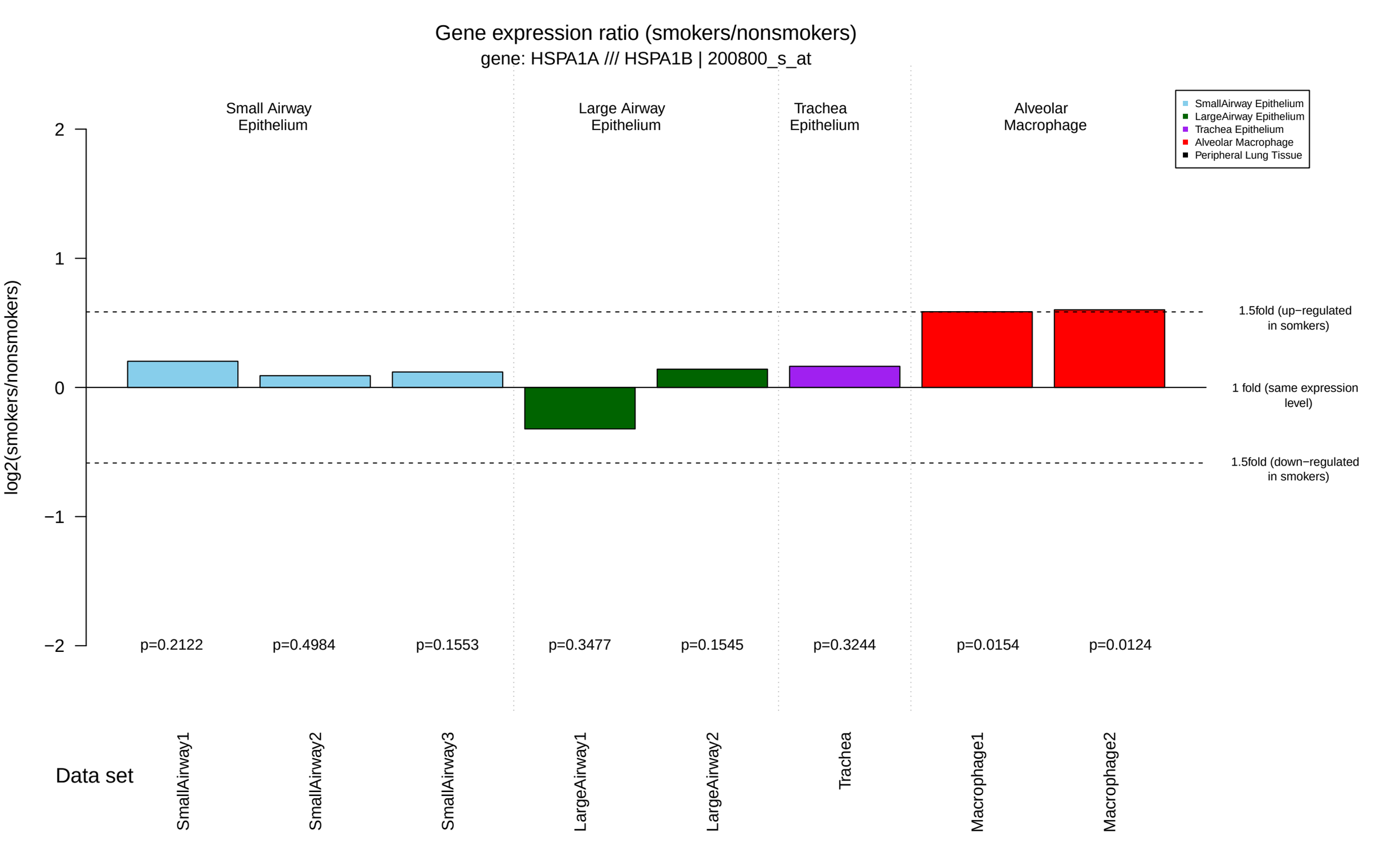
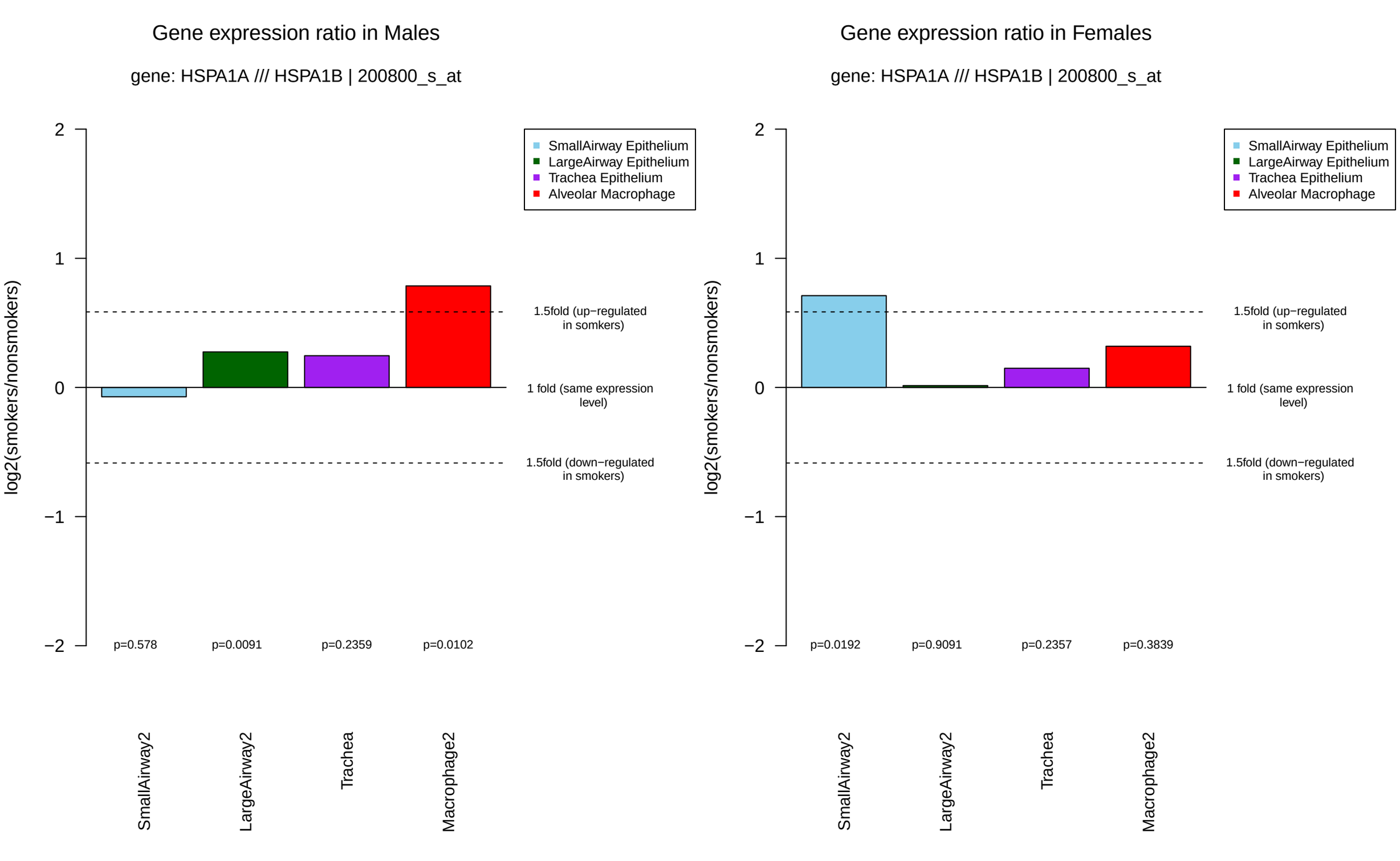
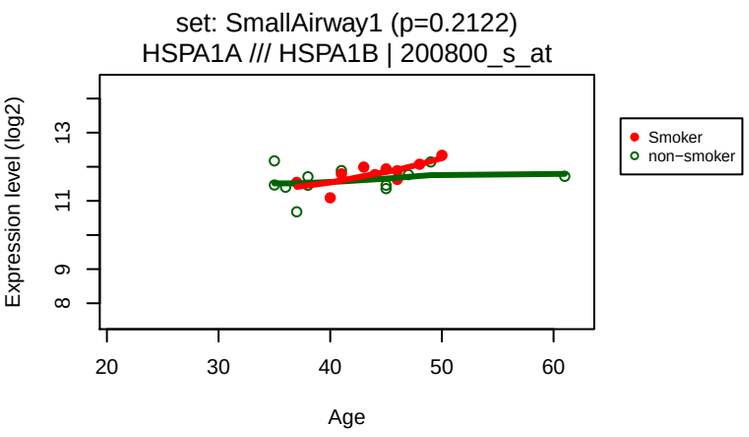
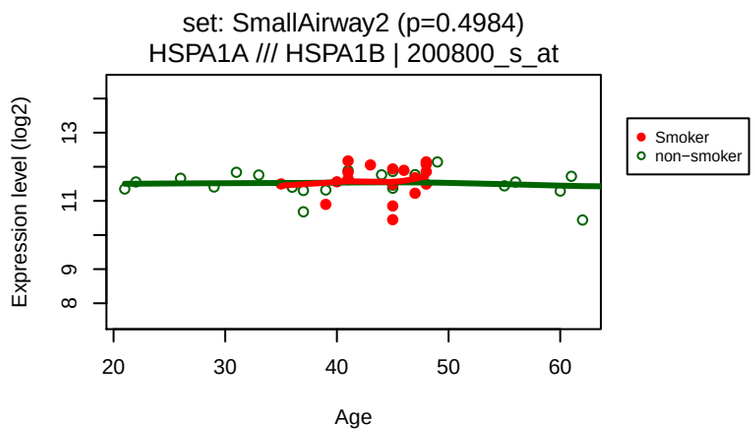
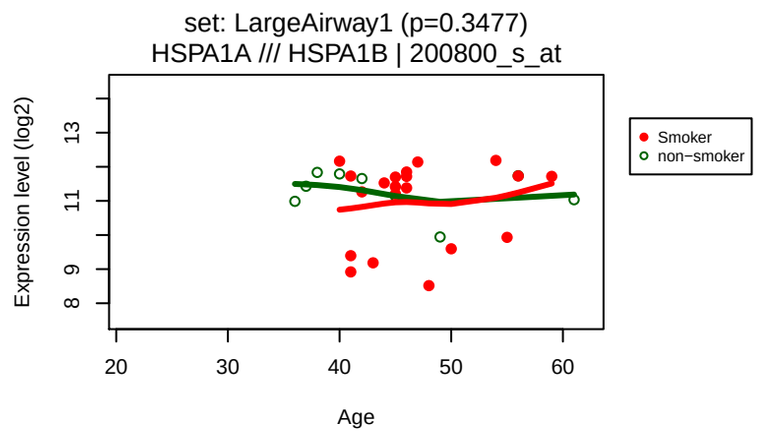

Gene expression of smokers and nonsmokers
gene: HSPA1A /// HSPA1B | 200800_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2122
p=0.4984
p=0.1553
p=0.3477
p=0.1545
p=0.3244
p=0.0154
p=0.0124
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

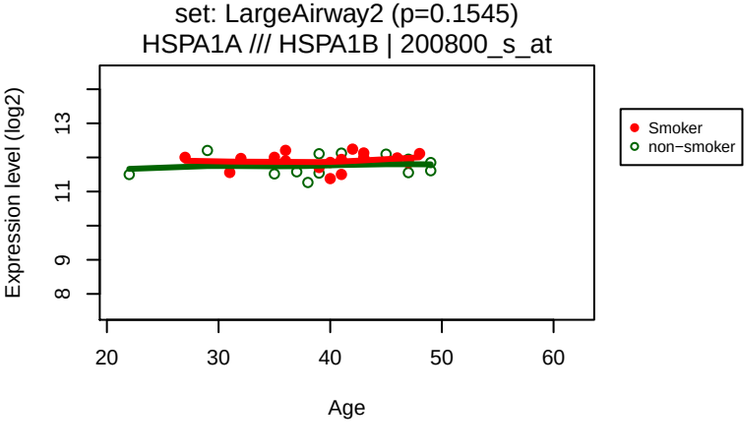

Gene expression ratio (smokers/nonsmokers)
gene: HSPA1A /// HSPA1B | 200800_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2122
p=0.4984
p=0.1553
p=0.3477
p=0.1545
p=0.3244
p=0.0154
p=0.0124
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

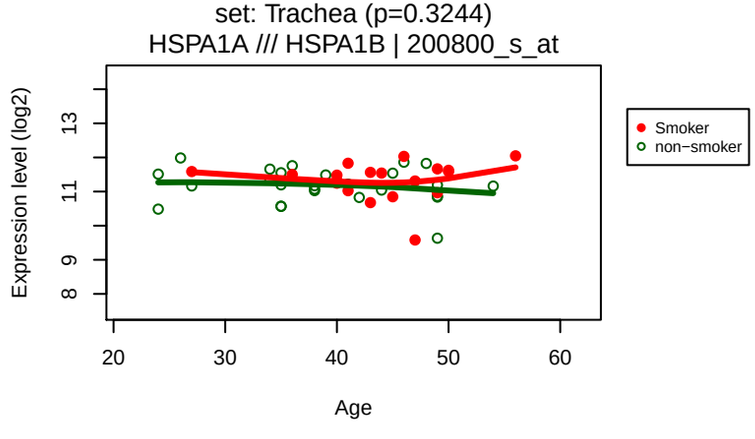

Gene expression ratio in Males
gene: HSPA1A /// HSPA1B | 200800_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.578
p=0.0091
p=0.2359
p=0.0102
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HSPA1A /// HSPA1B | 200800_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0192
p=0.9091
p=0.2357
p=0.3839
SmallAirway2
LargeAirway2
Trachea
Macrophage2

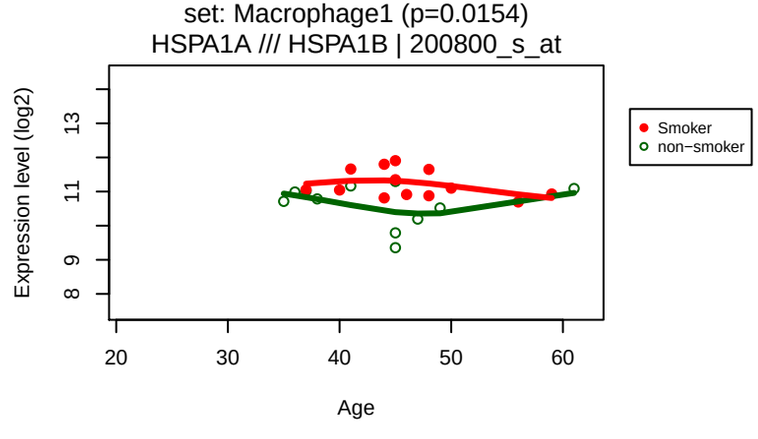

set: SmallAirway1 (p=0.2122)
HSPA1A /// HSPA1B | 200800_s_at
Expression level (log2)
8
9
11
13
20
30
40
50
60
Age
Smoker
non−smoker

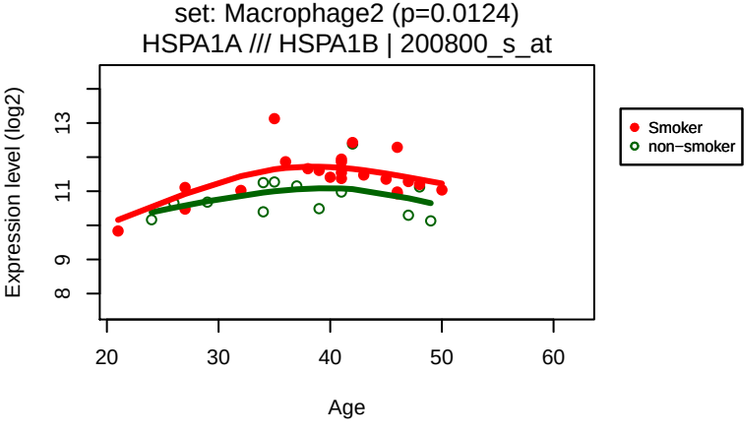

set: SmallAirway2 (p=0.4984)
HSPA1A /// HSPA1B | 200800_s_at
Expression level (log2)
8
9
11
13
20
30
40
50
60
Age
Smoker
non−smoker

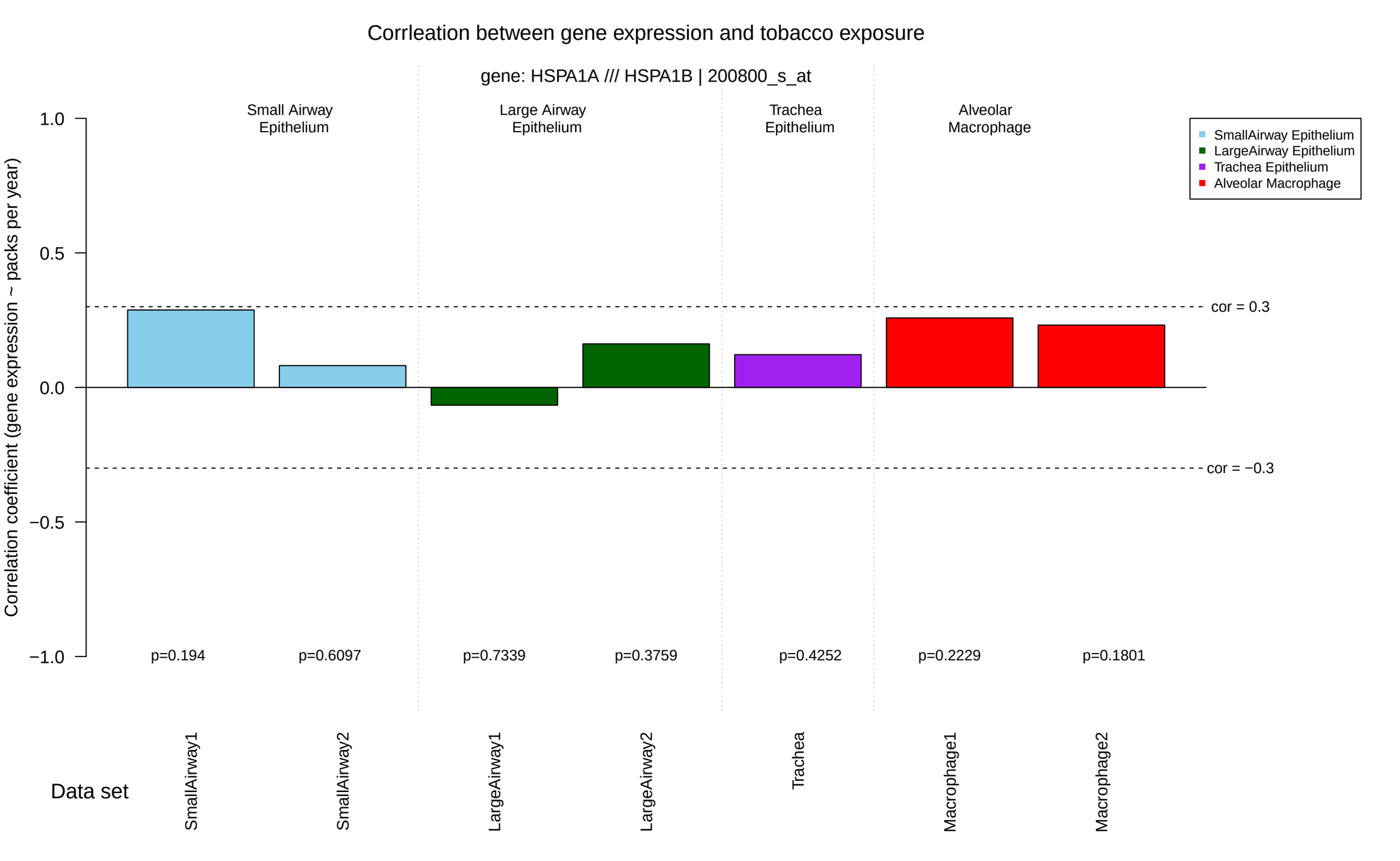

set: LargeAirway1 (p=0.3477)
HSPA1A /// HSPA1B | 200800_s_at
Expression level (log2)
8
9
11
13
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1545)
HSPA1A /// HSPA1B | 200800_s_at
Expression level (log2)
8
9
11
13
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.3244)
HSPA1A /// HSPA1B | 200800_s_at
Expression level (log2)
8
9
11
13
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0154)
HSPA1A /// HSPA1B | 200800_s_at
Expression level (log2)
8
9
11
13
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0124)
HSPA1A /// HSPA1B | 200800_s_at
Expression level (log2)
8
9
11
13
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HSPA1A /// HSPA1B | 200800_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.194
p=0.6097
p=0.7339
p=0.3759
p=0.4252
p=0.2229
p=0.1801
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2